

Study Guide for Ecosystems

This study guide provides examples for the topics that you might see on the final exam over the Ecosystems lectures. This is NOT an exhaustive list of all content that might be covered on the exam. Use your notes from lecture, the lecture slides and video on Canvas, and your homework questions to review all the topics that have been covered.

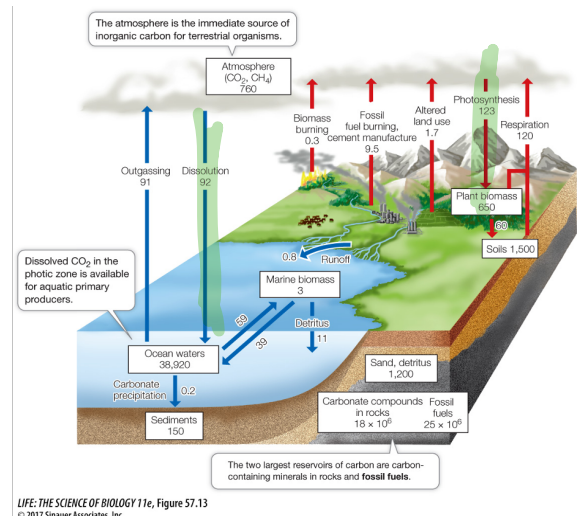
Flux = carbon movers

1. From the carbon cycle diagram below, which of the following is/are a process by which carbon can move out of the atmospheric pool? (Select all that apply)

- ☒ a. biomass burning
- ☒ b. fossil fuel burning
- ☒ c. altered land use
- ☒ d. **photosynthesis**
- ☒ e. respiration
- ☐ f. outgassing
- ☐ g. **dissolution**
- ☐ h. runoff

Read Question!

Arrow pointing down from Atmosphere ...



Sunlight = energy for photosyn; More light more food for eco!

2. Fill in the blanks: On land, net primary productivity is primarily driven by [blank1], while in the ocean, net primary productivity is primarily determined by [blank2].

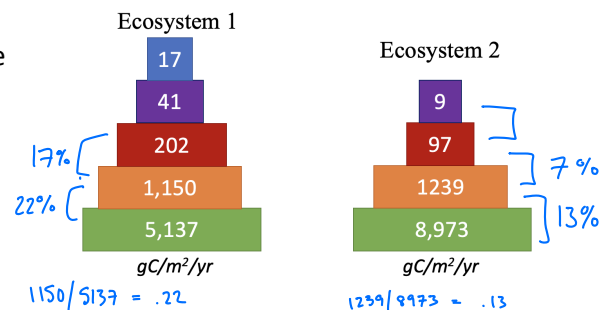
Nutrients = Builds Plant tissue; w/o it, even w/ light, plants cannot grow. Nitrogen, Phosphorus, iron, etc

Answer options: the concentration of greenhouse gasses in the atmosphere, the amount of solar radiation, the frequency of disturbance events, the amount of available nutrients, the species diversity of the plant community, average annual precipitation

3. You are studying the energy available at different trophic levels in two ecosystems and you develop the energy pyramids below to represent the energy availability at each trophic level in each ecosystem (in units of grams of carbon available per square-meter of habitat per year). Why is Ecosystem 1 capable of supporting five trophic levels, when Ecosystem 2 can only support four?

Idea: A lot of energy is lost going up trophic levels. More efficiency more energy reaches the top trophic levels. More efficiency = support more trophic levels

- ☒ a. Because Ecosystem 1 has higher efficiency of energy transfer across trophic levels than Ecosystem 2
- ☐ b. Because Ecosystem 1 has more energy entering the ecosystem through the producer trophic level than Ecosystem 2
- ☐ c. Because Ecosystem 1 follows Lindeman's Law and Ecosystem 2 does not
- ☐ d. Because Ecosystem 1 is a terrestrial ecosystem and Ecosystem 2 is an aquatic ecosystem



4. In class, we discussed the conditional interaction between plants and mycorrhizal fungi. What conditions determine whether this interaction is a mutualism or parasitism of the plant by the fungi?

- ☐ a. whether the plants are growing in high or low elevation ecosystems
- ☐ b. whether the two species occur sympatrically or allopatrically
- ☐ c. whether the ecosystem is experiencing primary or secondary succession
- ☐ d. whether the ecosystem is experiencing a top down or bottom up cascade
- ☒ e. whether the soil nutrient availability is high or low

Plant gives sugar to fungus via photosyn. Fungi gives nutrients + water from soil using its hyphae network.

*Low soil nutrients:
- Fungus helps plant
- Fungus gets sugar
- Mutualism*

*High soil nutrients:
- Fungus doesn't help
- Fungus still gets sugar
- Parasitism*

5. Some butterflies illustrate a form of mimicry, in which non-toxic species mimic the coloration patterns of toxic butterflies. In this type of mimicry, the non-toxic species benefits because predators assume it is as harmful to consume as the toxic species it mimics. When considering the ecological efficiency between the trophic levels of the non-toxic mimic and its predators, this form of mimicry most directly affects which energy efficiency step?

Butterfly

Bc ↑ reduces the total biomass the pred can eat. by tricking pred.

- a. Consumption efficiency: How much biomass is eaten
 b. Assimilation efficiency: How much biomass is digested / absorbed
 c. Production efficiency: How much absorbed energy turns into body mass.
 d. All efficiency steps are equally affected

Ecological Efficiency = CAP

6. Explain why positive interactions are more likely or common under "stressful" conditions. Draw a graph depicting how the strength of positive interactions changes along some environmental stress gradient (predation pressure, physical stress, etc). Ex) Plant + Mycorrhizal fungi: Low nutrient = help ea other

Ex) In high elevations, wax currant seedling grow under the shade of limber pines to avoid harsh cold, wind, sun conditions. Survival <20% → 40-80%.

7. Using a specific example of an interspecific interaction, describe how the same two species can interact to produce a range of outcomes, from positive (facilitative, mutualistic, etc) to negative (competitive, parasitic, etc). What components of the physical or biotic environment determine whether the interaction is positive or negative? Ex) Plant + Mycorrhizal fungi:

Low nutrient = mutual; high nutrient = parasitism or commensalism depending on how cost to plant (much sugar fungus uses)

8. Describe the differences between primary and secondary succession and give an example of the type of disturbance that would lead to each pattern of succession. Succession: Recover process after disturbance.

P: Starts from bare-ground w/o soil. Slow process (hundreds of years)

Ex) Glacier Retreat, lava flows, Sand dunes forming

S: Soil is present Ex) Forest fires, abandoned farms

9. What are the characteristics of typical early successional species and late successional species?

Early = pioneer = r-selected → quick + bulk reproduction. Ex) insects, weeds, bacteria (unstable env)

Intermediate: Shrubs, pines, tiny plants... Late = climax comm → K-select → Slow, few kids, stable env. (after ~500 years) Ex) Elephant, oak trees, Humans

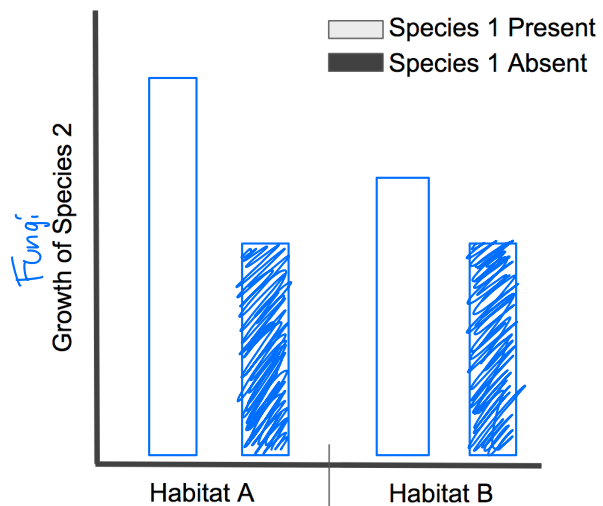
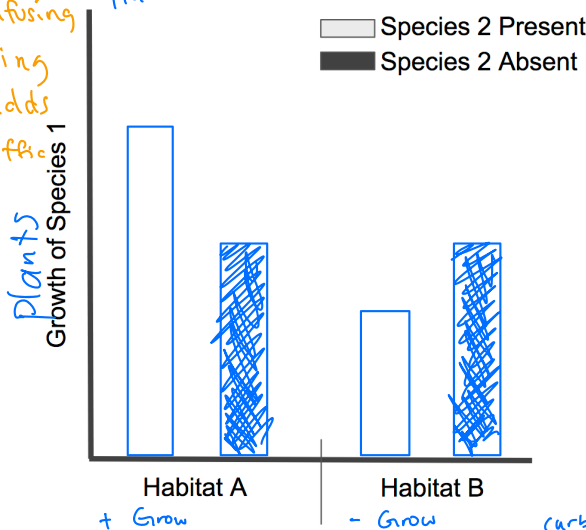
10. The graphs below show growth of Species 1 in Habitats A and B when Species 2 is present and absent (left graph) and the growth of Species 2 in Habitats A and B when Species 1 is present and absent (right graph). Complete the bar graphs with data that represent a conditional interaction.

Are there a lot of these? Confusing wording. Labeling + example adds a unnecessary diffc

Hab A = low nutrient

Hab B = high nutrient

any example where relation changes...



11. Ecosystem A has primary production of 1000 g C / m² / yr and ecological efficiency of 10%.

Underman's Law: Ecosystem B has primary production of 300 g C / m² / yr and an ecological efficiency of 25%.

Tertiary Con (the rest)
 ↑
 Secondary Con (carn)
 ↑
 Primary Con (Herb)
 ↑
 Primary Pro (Plants)

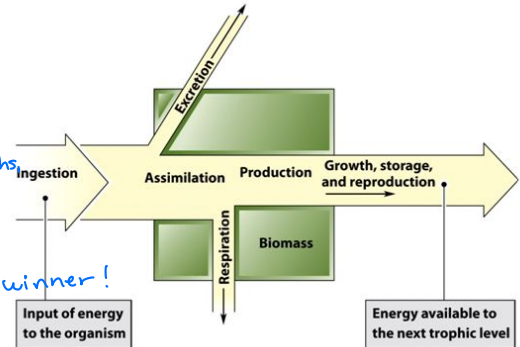
- a. Which ecosystem will have more production at the secondary consumer (carnivore) trophic level? The one w high efficiency = Ecosystem B = $300 \cdot .25 \cdot .25 = 18.75$ (A = $1000 \cdot 0.1 \cdot 0.1 = 10$)
 b. Which ecosystem is more likely to support an endothermic primary consumer (herbivore)? The one with the highest at second level = Ecosystem A ($1000 \cdot 0.1 = 100$)
 Eco B ($300 \cdot .25 = 75$) loses.

carbon stored in biomass

- c. If a trophic level requires at least $1 \text{ gC/m}^2/\text{yr}$ in order to exist, how many trophic levels can each of these ecosystems support? Given your answer to part c, which do you think has a larger impact on the energy available at upper trophic levels: primary productivity or efficiency of energy movement across trophic levels?

B is more much efficient so there's more available gC/m^2 at each level compared to A. Efficiency > Starting amount

12. Describe how the following terms relate to one another: consumption efficiency, assimilation efficiency, production efficiency, and ecological efficiency. Each are steps of energy movement through the trophic levels. How much biomass is eaten, absorbed, turned into new body mass. Putting these together describes ecological efficiency.
13. The diagram at right describes how energy moves through a trophic level. Recall the definitions of the different types of efficiencies to answer the questions below



- . Would you expect the assimilation efficiency to be higher for a **carnivore** or an herbivore?

Digesting plant cell walls is difficult for most organisms (cows multiple stomachs, chewing cud)

- a. Would you expect the production efficiency to be higher for an **ectotherm** or an endotherm?

Respiration rate determines prod efficiency.

Endo = higher respiration rate to maintain body temps. = less efficient. Ecto = winner!

- b. Would you expect the consumption efficiency of primary consumers to be higher in a forest

or a grassland? Grass. Consumption efficiency is eaten. Since most organism cannot eat wood the grassland has more available biomass. Grassland wins!

- c. Which would you expect to have higher production efficiency: a **C3 plant** or a C4 plant?

C3: Use less CO_2 for energy but susceptible to dehydration. C4 stores CO_2 in special cells for better water retention, but cost more energy.

14. Why is terrestrial net primary productivity higher in the tropics, but marine net primary productivity is higher in coastal areas?

On land NPP is controlled by temps. Warm places (tropics) have high productivity, esp when accompanied w/ rain.

NPP depends on nutrients. Coastal areas are very productive because runoff (from fertilizer/waste) + upwelling of currents bringing the nutrients. Gulf of Mexico get their nutrients from Mississippi river.

15. You may have heard it argued that being a vegetarian is more energy efficient than eating meat.

Use your knowledge of energy transfer across trophic levels to describe whether you accept or reject this argument. Plants take more energy to break down so herbivores have lower assimilation eff.

Consequently herbivore production eff is low so less biomass for next trophic level. Carnivores breakdown tissue easier = more eff overall.

16. What are dead zones and why do they commonly form at the mouth of large rivers?

Areas w/ too much nutrients create algal blooms; when they die the bacteria that decomposes them take up all oxygen in that area leaving little for organisms to survive. DZ form near river mouth bc they carry large nutrient runoff from farm/waste. Mississippi → Gulf of Mexico.

17. The diagram of the hydrologic cycle states that groundwater plays a small role in the global water cycle because it is 'locked in' to rock and soil layers. How is groundwater important for humans?

How does the disconnect of groundwater from the global water cycle impact our future?

Good for midwest farms that rely on Ogalla Aquifer for: corn, wheat, cotton, beef. Unfortun., it recharges slowly; we're pumping it out faster than its refilling. Gw is becoming harder/expensive to access.

18. In class, Laci stated that diagrams of energy across trophic levels must always have a pyramid shape (most energy in the producers, decreasing at each level up to the top predators), but that diagrams of biomass or numbers of individuals in each trophic level can be a variety of shapes.

Describe why energy diagrams must be a pyramid, and describe examples of ecosystems that may have biomass or number of individuals that are not pyramid-shaped. It takes the pyramid shape...

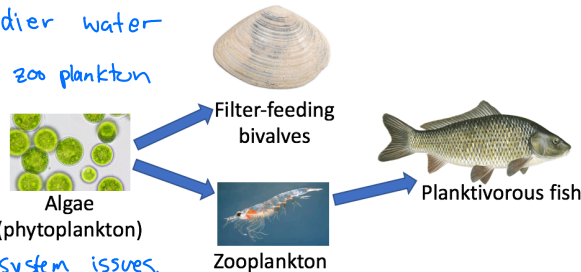
TC+...
TC
SC
PC
PP

.. bc the PP level is the only biomass producer. Every level above, energy is lost through respiration/pooping. Ex) Phytoplankton reprod so fast they

19. You are an environmental consultant hired by a town adjacent to a small lake in the Sierras. The figure below describes what you know about the food web of the lake. The town is economically depressed, and residents have proposed two ideas for stimulating the local economy (1) even if their biomass is

introduce northern Pike (a fish that only eats other fish) into the system to attract sport fishers (and their money) to the area and (2) harvest filter-feeding bivalves from the lake for food (they are quite a delicacy and fetch a hefty price). The town wants you to predict for them the consequences of each proposal for the health of their lake, long renowned for its clear waters.

- Harvest Bivalves = increase algae = cloudier water
- Intro Pikes = Reduce fish species that eat zooplankton = increase zooplankton size since less of those fish are around = less algae since zooplankton eat them = clear water
- Final: Adding a pred is risky bc invasive species are unpredictable and can cause ecosystem issues



Differ: Initial trophic level

- Perturbations = Disruptions in eco that ripple trophic levels.
20. What is the difference between a top-down trophic cascade and a bottom-up trophic cascade? Give an example of each.
- TD: Starts @ Pred (tertiary consumer); affects down.
Ex) Orcas → sea otters → sea urchins → kelp
- BU: Starts @ producers/nutrients and affects higher levels.
Ex) Mississippi runoff into Gulf of M. → Dead zones (algal bloom → oxygen loss) → dead fish...
21. In class, Laci referred to the Ogallala Aquifer as 'fossil water.' How are aquifers similar to other 'fossil' resources like oil and coal? Both come from underground & slow to replenish.
- Aquifers are stored water and are not formed from organisms; water ≠ coal or oil.

22. Use the diagram below of the global nitrogen cycle to answer the following questions. Note: for the nutrient cycle diagrams, I do **NOT** want you to memorize pools and fluxes, I **DO** want you to be able to read and interpret a cycle diagram

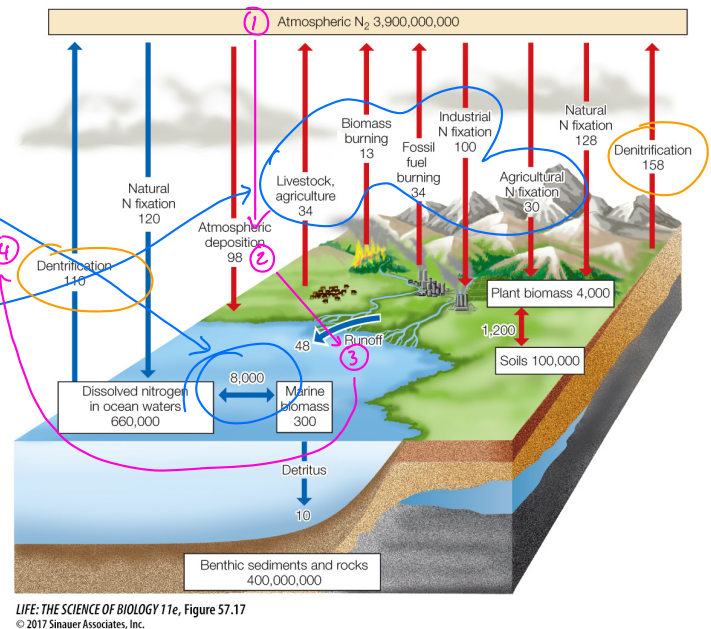
a. What is the largest flux? What is the largest pool?

Pool = Atmosphere. Flux =

b. Which of the fluxes are anthropogenic (caused by humans)? How do these fluxes compare in magnitude to the natural fluxes?

Human Atmosphere = $34 + 13 + 34 = 81$
Denitrification = $110 + 158 = 268$

c. Describe a pathway by which a molecule of nitrogen in the atmosphere could move through both terrestrial and aquatic pools and then return to the atmosphere.



Atmosphere →

Deposition into land → Runoff into ocean from river → Denitrification into Atmosphere